

Statistics Hybrid Practice Set – S5 (L2)

Topics

- Population
- Sample
- Mean
- Median
- Mode
- Range
- **Variance (New)**

Instructions

- Solve on **paper first**.
- Then solve the **same question in Excel**.
- Write the Excel formula you used.
- No calculator.

Q1. Identify Population and Sample

A retail company has **6,000 customers**. It surveys **250 customers**.

Paper Task

- Population = ?
- Sample = ?

Excel Task

Total Customers	Surveyed Customers
6000	250

(No formula required.)

Q2. Calculate the Mean

Value
11
15
19
23
27

Paper Task

- Calculate the Mean.

Excel Task

- Calculate the Mean.
 - Write the Excel formula.
-

Q3. Find the Median

Value
35
15
25
10
20

Paper Task

- Find the Median.

Excel Task

- Calculate the Median.
 - Write the Excel formula.
-

Q4. Find the Mode

Value
9
11
11
13
15
17

Paper Task

- Find the Mode.

Excel Task

- Calculate the Mode.
 - Write the Excel formula.
-

Q5. Calculate the Range

Value
22
35
18
41
29

Paper Task

- Calculate the Range.

Excel Task

- Calculate the Range.
 - Write the Excel formula.
-

★ New Topic – Variance

Formula (Paper)

$$\text{Variance} = \frac{\sum (x - \bar{x})^2}{n}$$

Note: In this practice set, always use **Population Variance**.

Q6. Calculate the Population Variance

Value
2
4
6
8
10

Paper Task

1. Find the Mean.
2. Find each deviation $(x - \bar{x})$.
3. Square each deviation.
4. Find the Population Variance.

Excel Task

- Calculate the Population Variance.
 - Write the Excel formula.
-

Q7. Calculate the Population Variance

Value
5
5
5
5
5

Paper Task

- Calculate the Population Variance.

Excel Task

- Calculate the Population Variance.
 - Write the Excel formula.
-

Q8. Interpretation

Dataset A

Value
20
21
22
23
24

Dataset B

Value
5
15
25
35
45

Paper Task

1. Which dataset has the **higher variance**?
2. Why?

Excel Task

Calculate the Population Variance for both datasets.

Q9. Real Data Analyst Scenario

Two stores have the **same average daily sales**.

Store A's sales are almost the same every day.

Store B's sales change a lot every day.

Which store has the **higher variance**?

A. Store A

B. Store B

Explain your answer in one sentence.

Q10. Excel Formula Challenge

Write the Excel formulas for:

1. Mean
2. Median
3. Mode
4. Range
5. **Population Variance**

Use the range **B2:B6**.